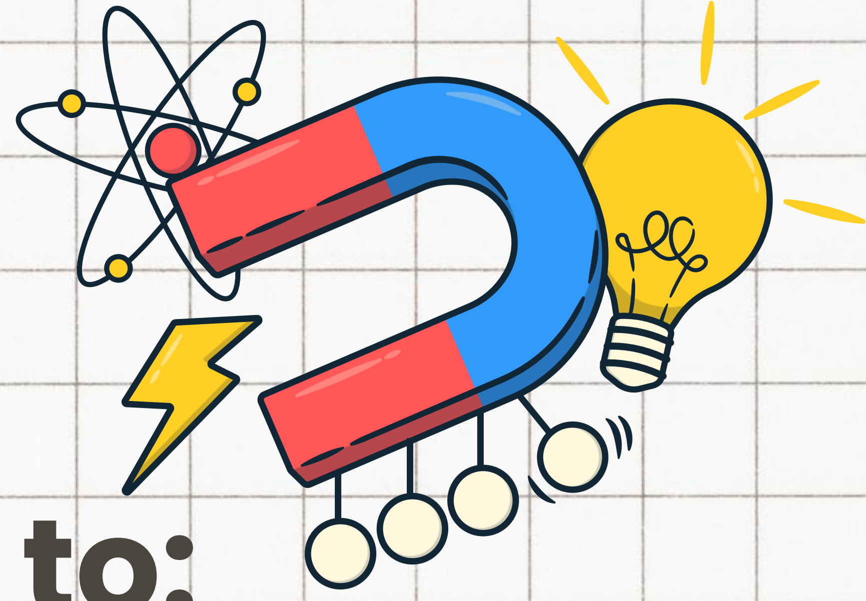
GENERAL SCIENCE
WEEK 4

The Physics of Fluids

NAME OF TEACHER

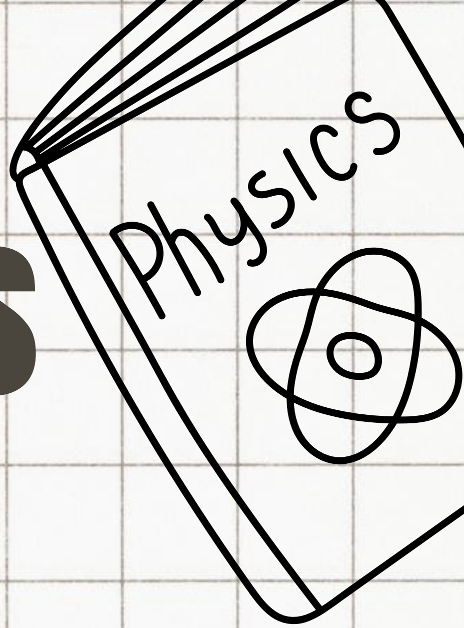
DAY 1: LEARNING OBJECTIVES



At the end o f the lesson, I should be able to:

- Define what a hydraulic system is and identify its fundamental components
- Identify how hydraulic systems are utilized in everyday life
- Explain the relationship between Pascal's Principle and hydraulics

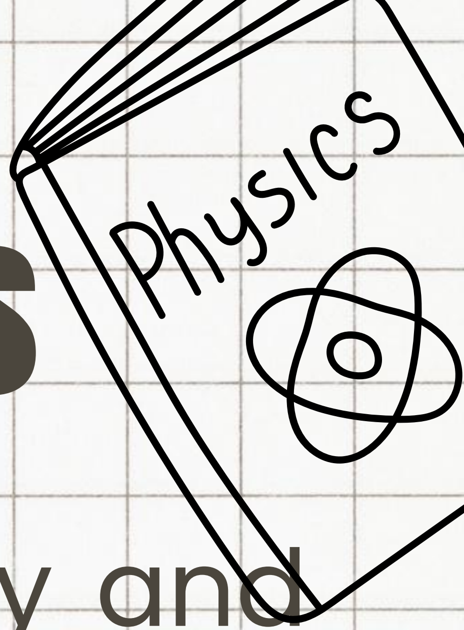
The Sand VS Nails



QUESTIONS:

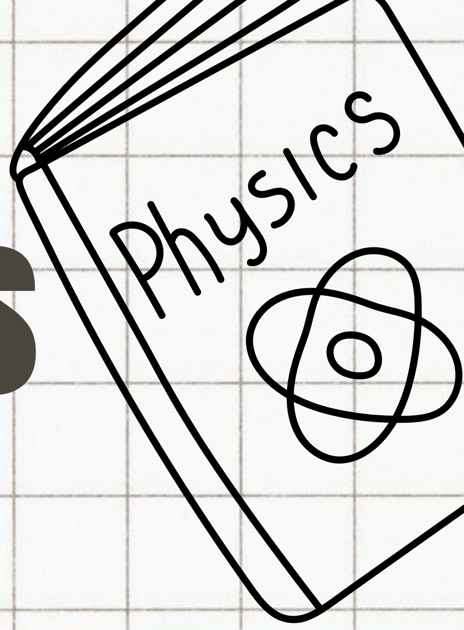
1. Compare the two images. What is the main difference you notice in how the surface (sand vs. nails) reacts to the person's step?
2. Even though the person's weight (force) stays the same, why do you think the results are so completely different?

The Sand VS Nails



- **Sand:** The foot sinks deeply and easily displaces the grains.
- **Bed of Nails:** The surface supports the weight without piercing the skin.
- **Key Contrast:** One surface yields completely, while the other resists and supports.

The Sand VS Nails



- **It's all about surface area.**

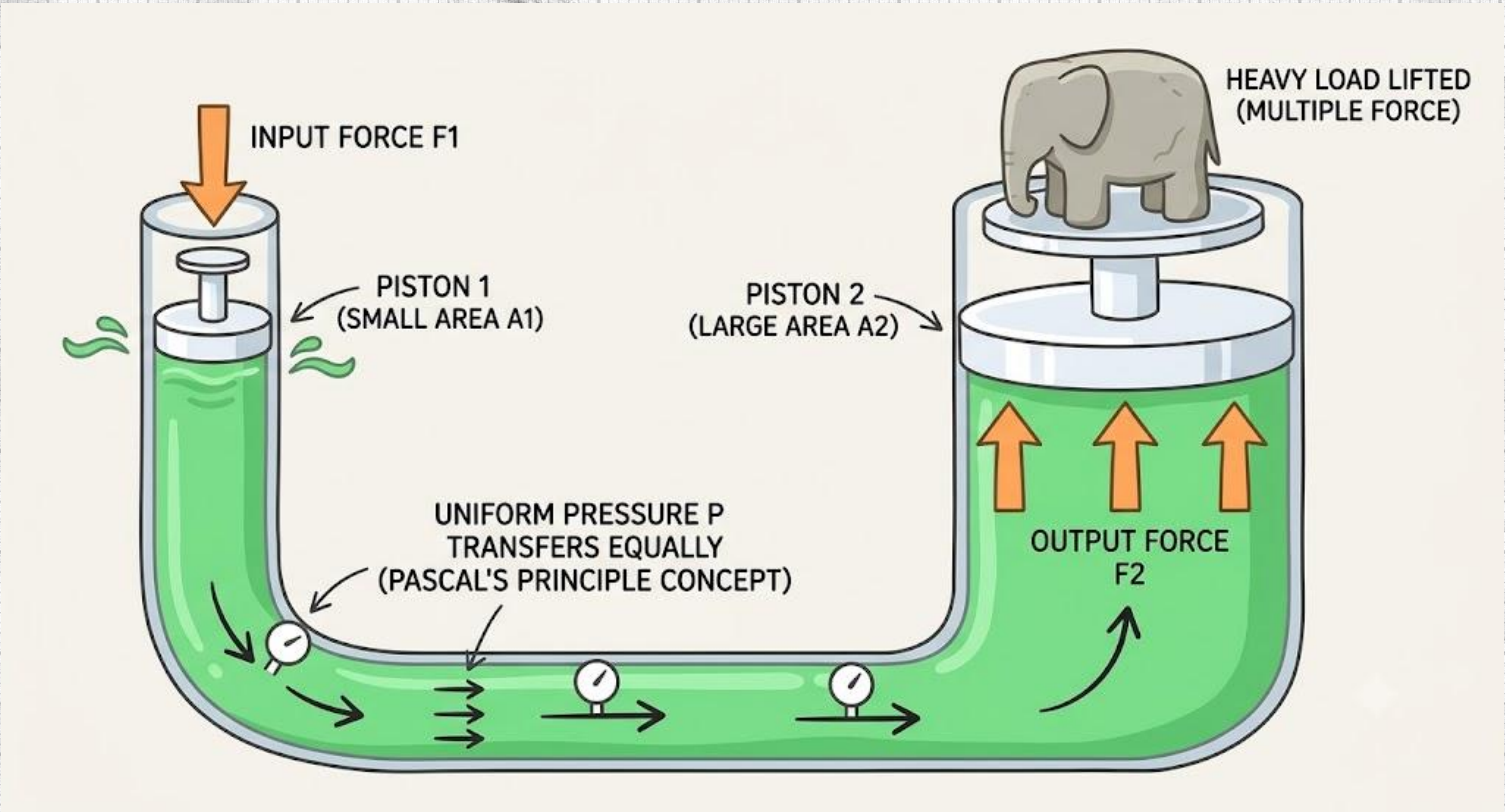
* **Sand:** The weight is concentrated only on the area of a foot.

* **Bed of Nails:** Hundreds of nail points multiply the total contact area.

- **The Takeaway:** Spreading the same weight over a larger area reduces the pressure at any single point

- We see how solid surfaces like sand and nails handle force. But what happens if we trap a fluid—like water or oil—underneath that person's foot instead? Unlike sand, which just moves out of the way, or nails, which stay rigid, ***a trapped fluid behaves in a completely unique way.***

The Piston Analogy

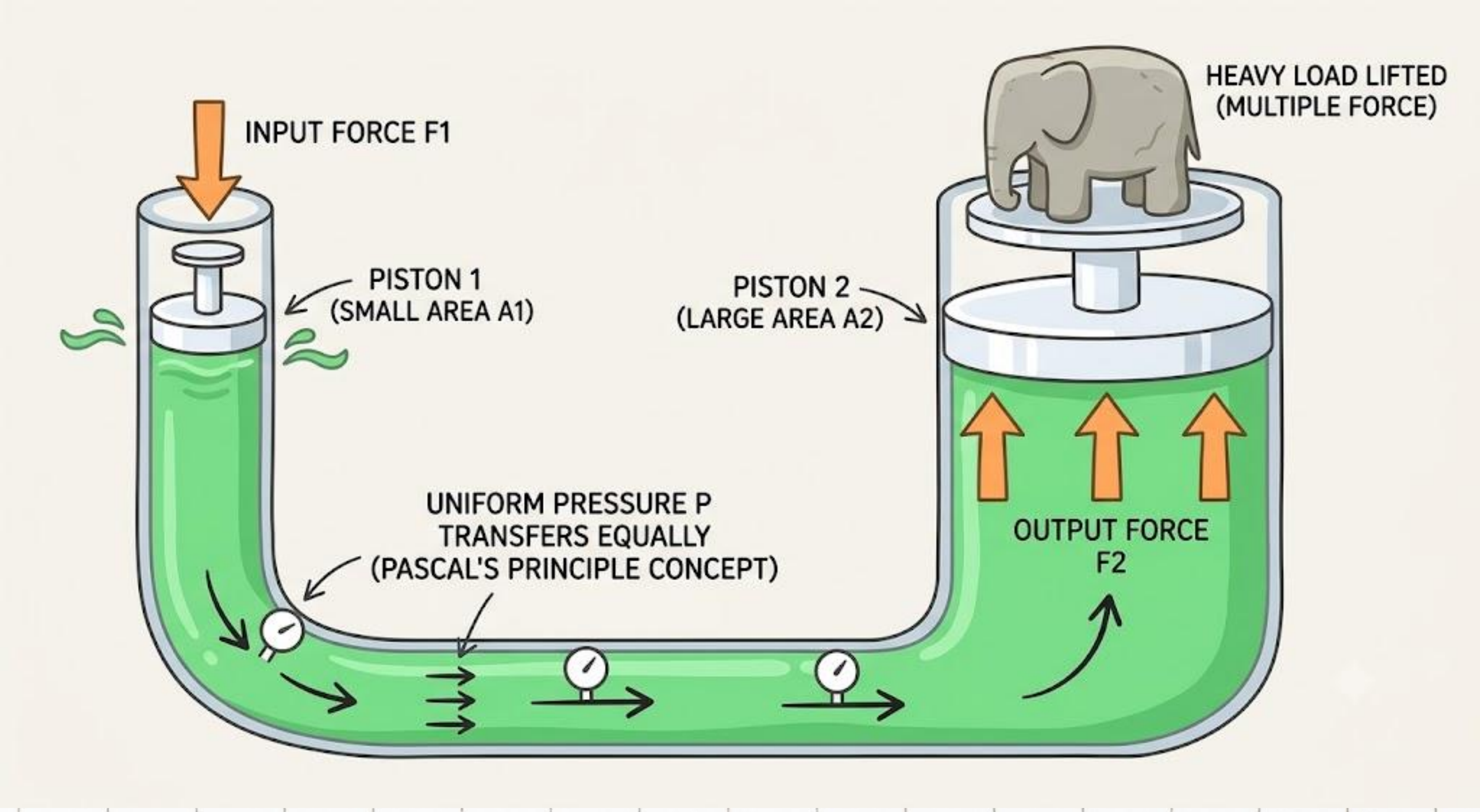


Contrast Solid vs. Fluid Response

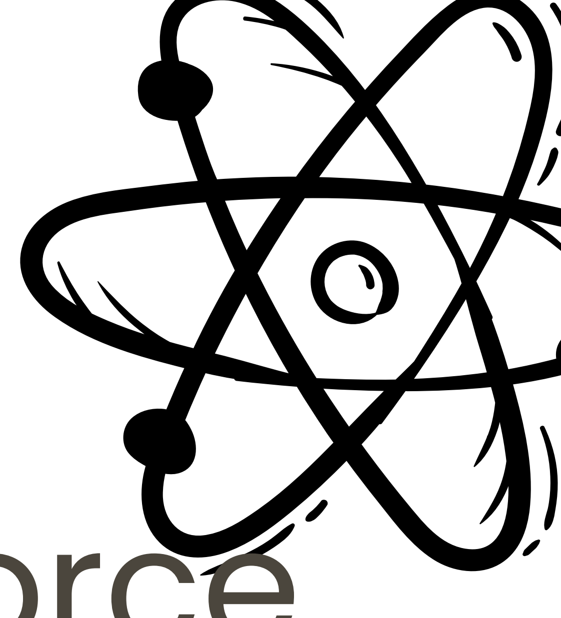
- **Solid Response (Nails/Sand):** The force travels in one direction, straight down into the ground.
- **Fluid Response:** The force turns into pressure that shoots out in every single direction at once with equal intensity.

The Piston Analogy

Since we know the fluid pushes equally everywhere, what if we give it a way out? What if we connect this container to a second, wider container with its own piston?



Because the fluid response distributes pressure evenly, the fluid carries that exact same pressure over to the wider side, pushing up on a larger area with a massively multiplied force. That is a **HYDRAULIC SYSTEM.**



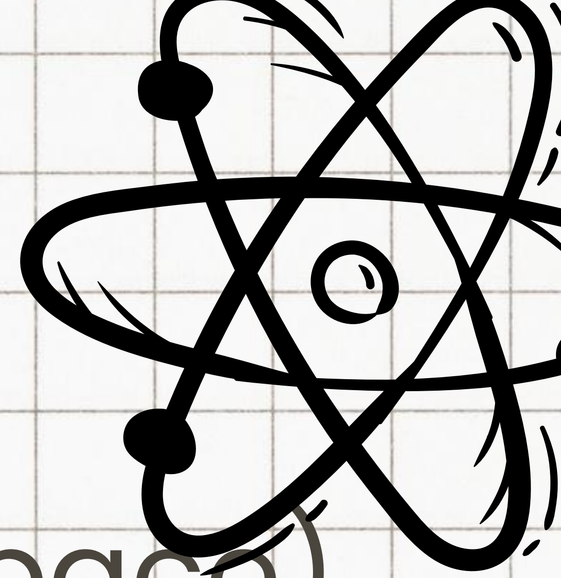
PRESSURE (P):

The measure of how concentrated a force is over a specific area.

2 IMPORTANT VARIABLES

- **Force (*F*):** The total downward push or weight. (This stayed constant whether stepping on sand or nails).
- **Area (*A*):** The total contact surface. (This changed from a single foot to hundreds of nail points)

The Logic Behind Equation



- If we increase **Force** (push harder on the same space)

→Pressure goes **UP**. (Force belongs on top of the fraction).

- If we increase **Area** (spread the same weight over more space) → Pressure goes **DOWN**. (Area belongs on the bottom of the fraction)

Formula:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

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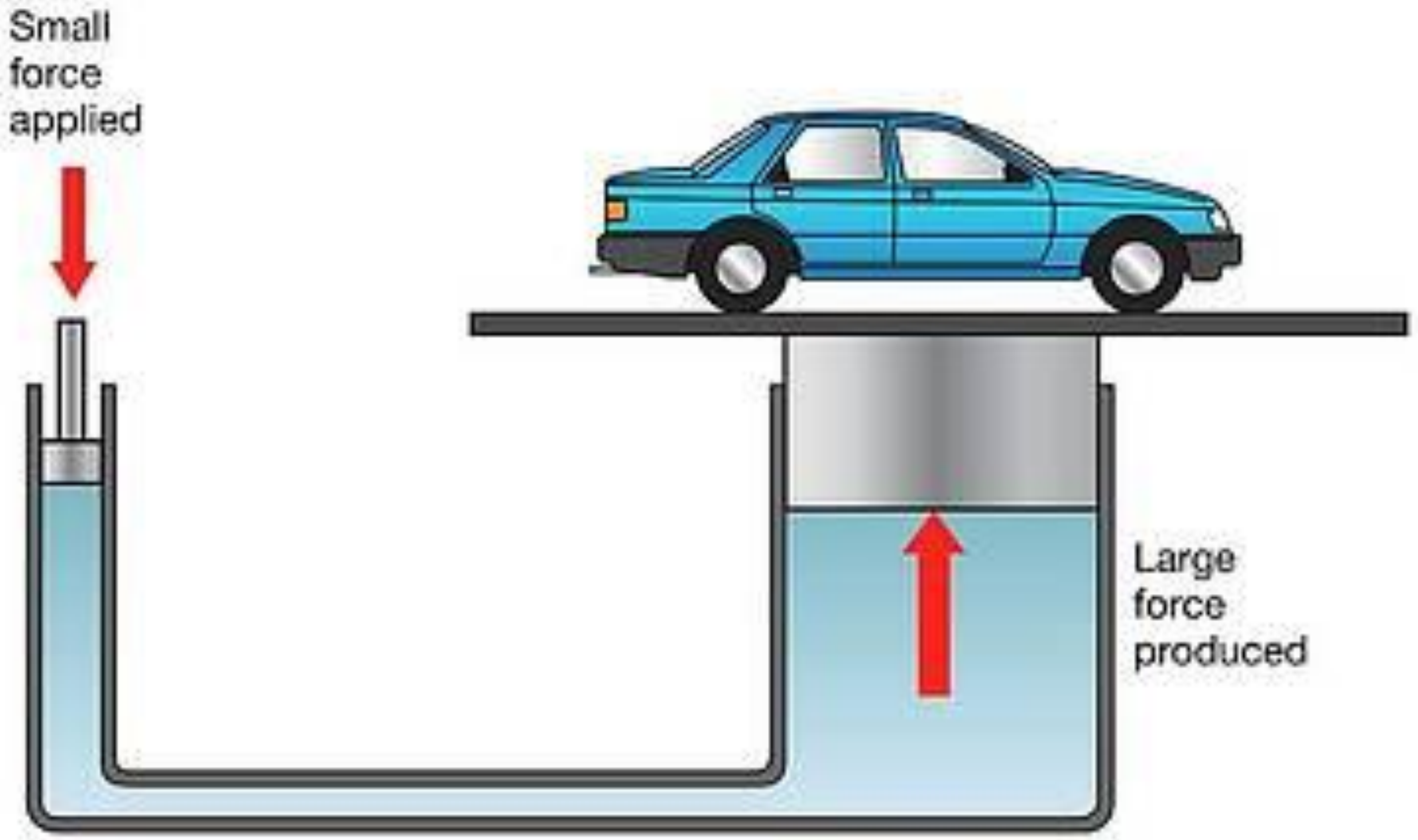
$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

- P = Pressure → Measured in Newtons per square meter (Nm^2), officially called the Pascal (Pa)
- F = Force → Measured in Newtons (N)
- A = Area -> Measured in Square Meters (m^2)

Physics

HYDRAULIC SYSTEM

Hydraulic system is a mechanical network that uses confined, incompressible fluids to transmit and multiply force. It allows a small input force to lift an incredibly heavy load, acting as a "*mechanical lever*" powered by liquid.



The Foundation: Pascal's Principle

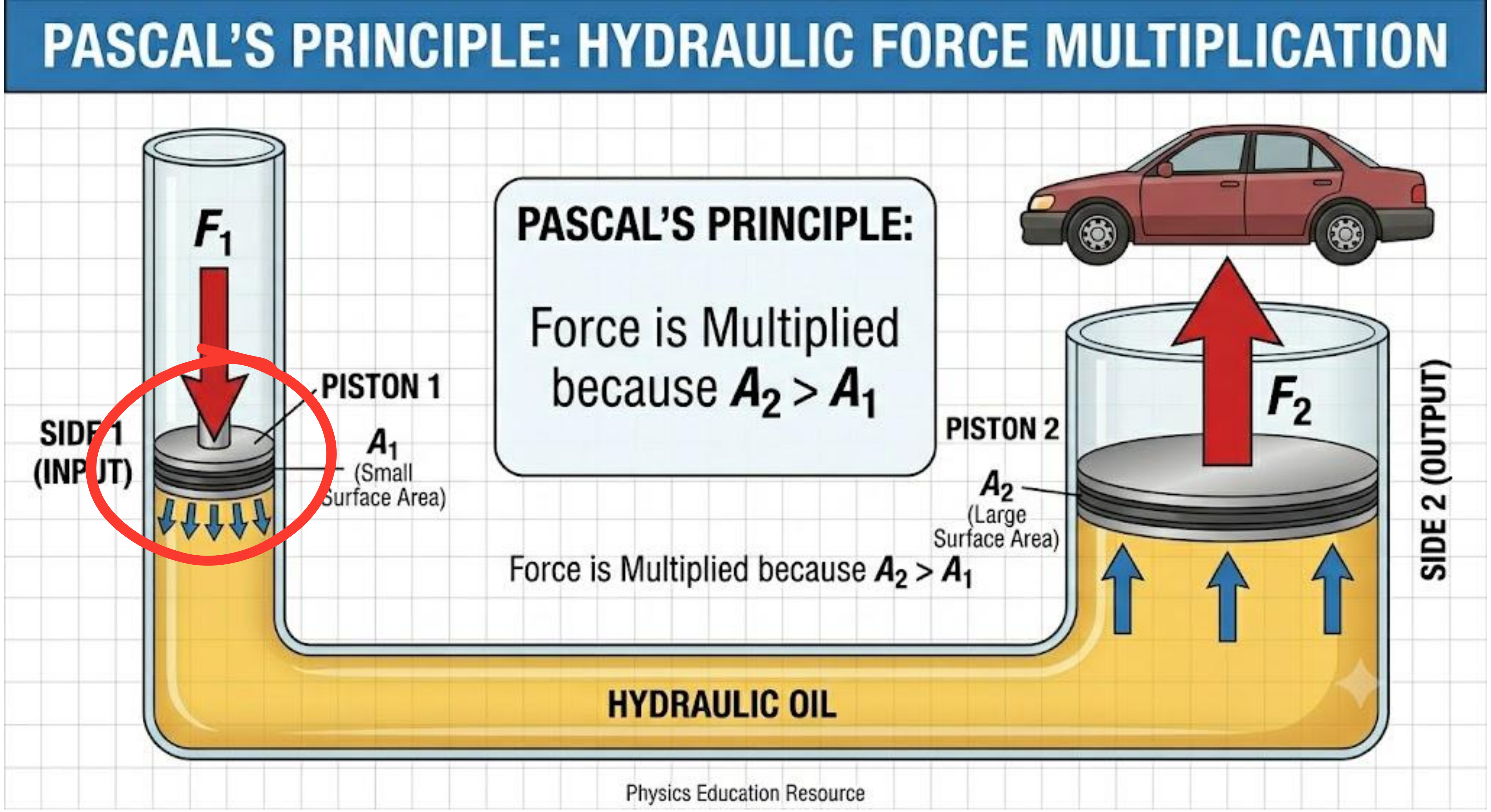
- **The Law (Blaise Pascal, 1647):** Pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.

BLAISE PASCAL, 1647

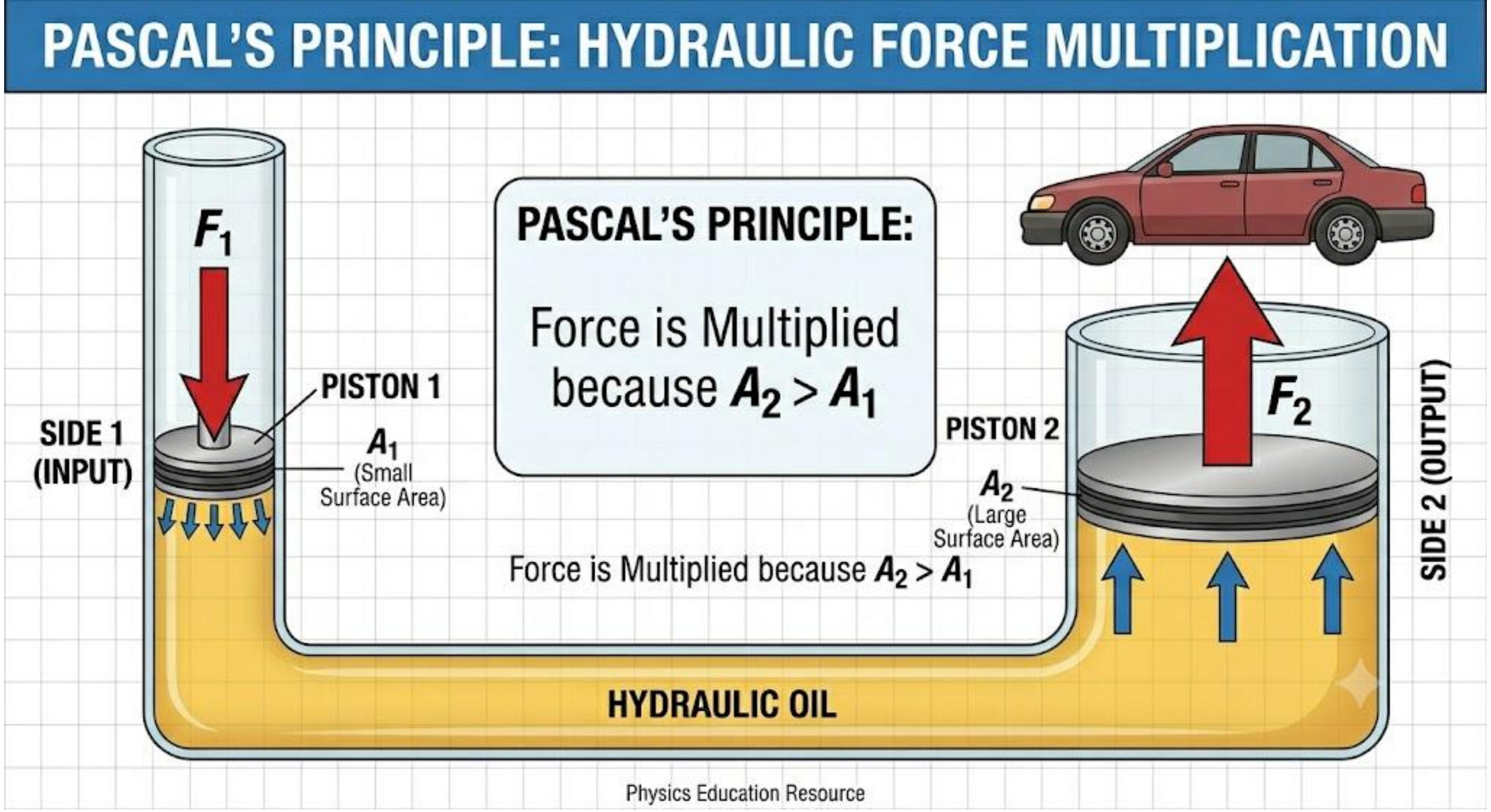
The Foundation: Pascal's Principle

- **The Rule of Liquids:** Unlike gases (which compress easily), liquids are incompressible. If you push down on trapped fluid, its molecules cannot squeeze closer together. Instead, they pass that push along instantly in all directions.

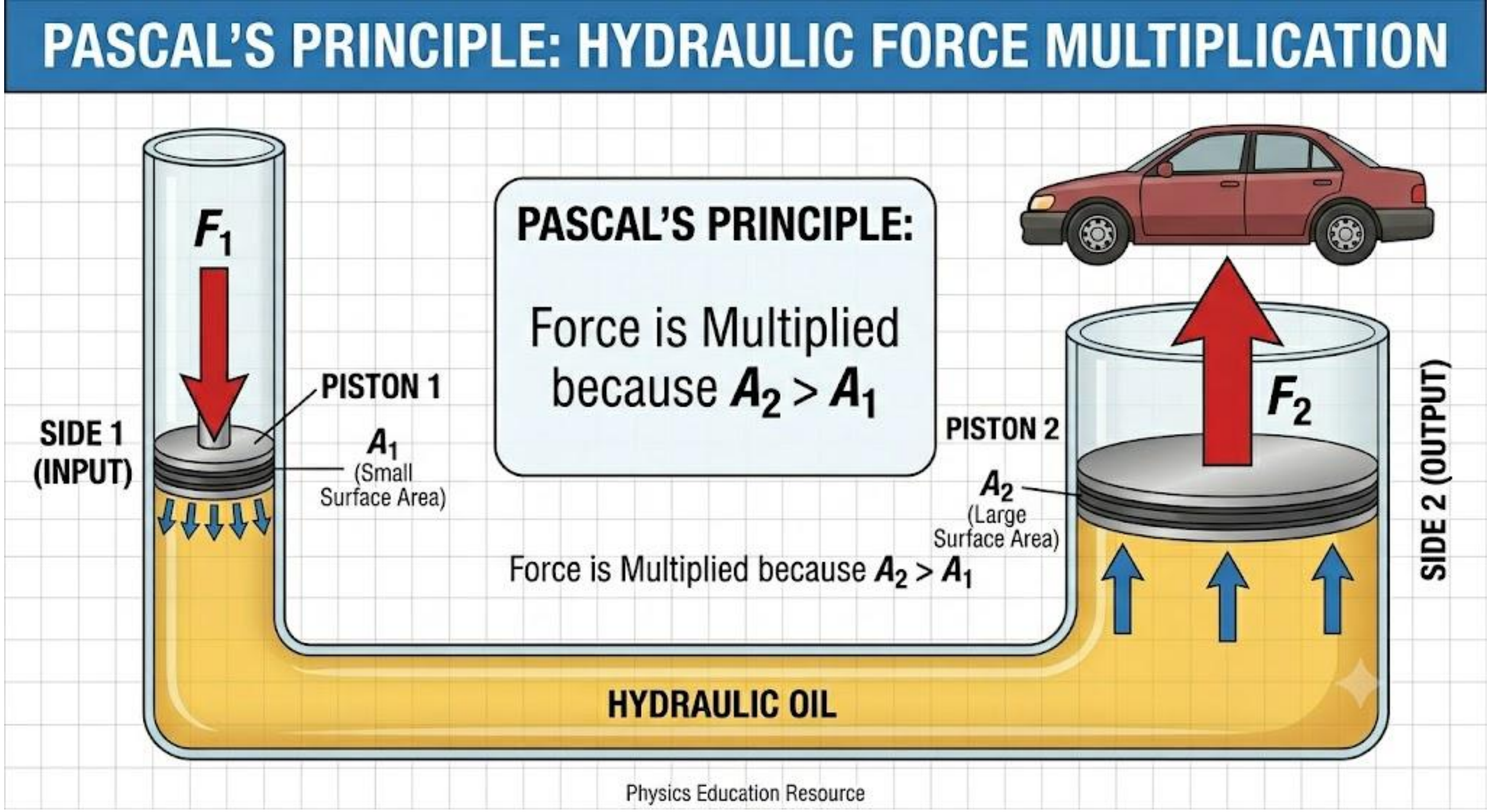
BLAISE PASCAL, 1647



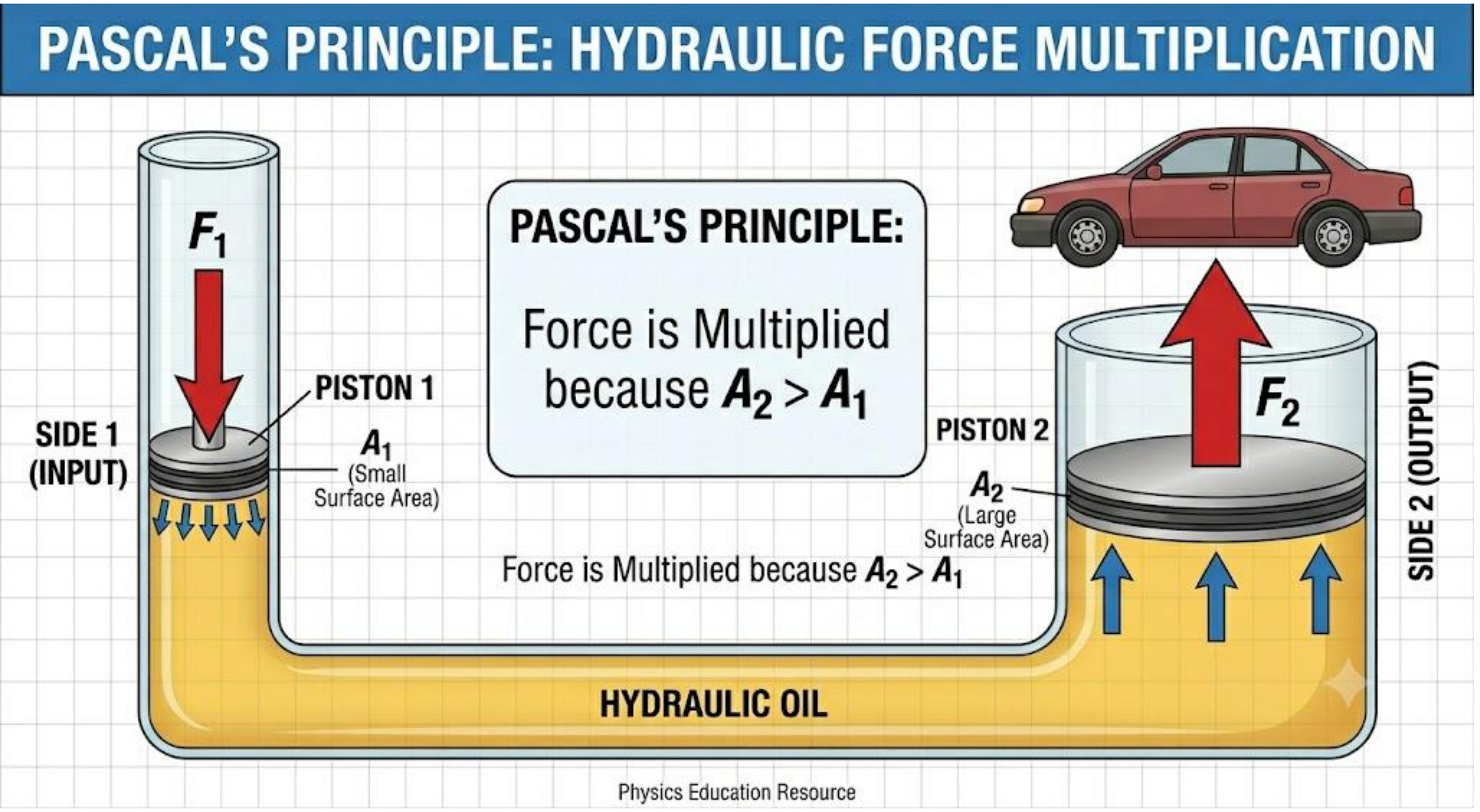
On the Left Side (*The Input*): We have a tiny piston. Because it's small, it doesn't take much effort to push it down. We apply a small force (F_1) over a small area (A_1)



The Magic in the Middle: When you push that left side down, you create fluid pressure. Because liquids cannot be squeezed or compressed, that pressure travels through the oil and pushes against every single inch of the tube with the exact same strength.



The Catch: You don't get this power for free. Because the left side is so small, you have to push it down a long distance just to make the heavy right side lift up a tiny fraction of an inch.



Hydraulics do not create free energy; they are a trade-off. You are multiplying force by sacrificing distance. The system works because the pressure (P) remains exactly the same on both sides of the liquid. Since the pressure is identical, making the second piston 10 times larger automatically makes the upward lifting force 10 times stronger .

HYDRAULIC SYSTEMS IN EVERYDAY LIFE

1. Transportation and Travel

Car Brake Systems

Every time you press your car's brake pedal, you are using a hydraulic system.

- **How it works:** Your foot pushes a small piston in the master cylinder. This creates pressure in the brake fluid, which travels through tubes to all four wheels. There, the fluid pushes against larger pistons that clamp the brake pads tightly against the spinning wheels.
- **The Physics:** A light tap of your foot is multiplied into enough force to stop a two-ton vehicle traveling at highway speeds.

1. Transportation and Travel

Heavy Construction Machinery (Excavators & Bulldozers)

If you watch a construction crane, excavator, or dump truck work, you will see shiny metal rods sliding in and out of large metal sleeves. These are hydraulic cylinders.

- **How it works:** High-pressure oil is pumped into these cylinders, pushing against large internal pistons.
- **The Physics:** This fluid response allows a single machine to smoothly scoop up metric tons of dirt, crush concrete, and lift steel beams that would break standard mechanical gears.

1. Transportation and Travel

Aircraft Flight Controls

Commercial airplanes fly through high-velocity winds that put immense pressure on their wings. A pilot could never physically muscle the wing flaps (ailerons and elevators) against that wind resistance.

- **How it works:** The pilot's cockpit controls send signals to powerful hydraulic pumps.
- **The Physics:** These pumps force fluid into actuators on the wings, multiplying the pilot's input force to easily tilt the massive flaps against rushing air.

2. Commercial and Public Spaces

Passenger Elevators

Many low-to-medium-rise elevators (like those in three-story hotels or office buildings) do not use overhead cables and pulleys. Instead, they are pushed from below.

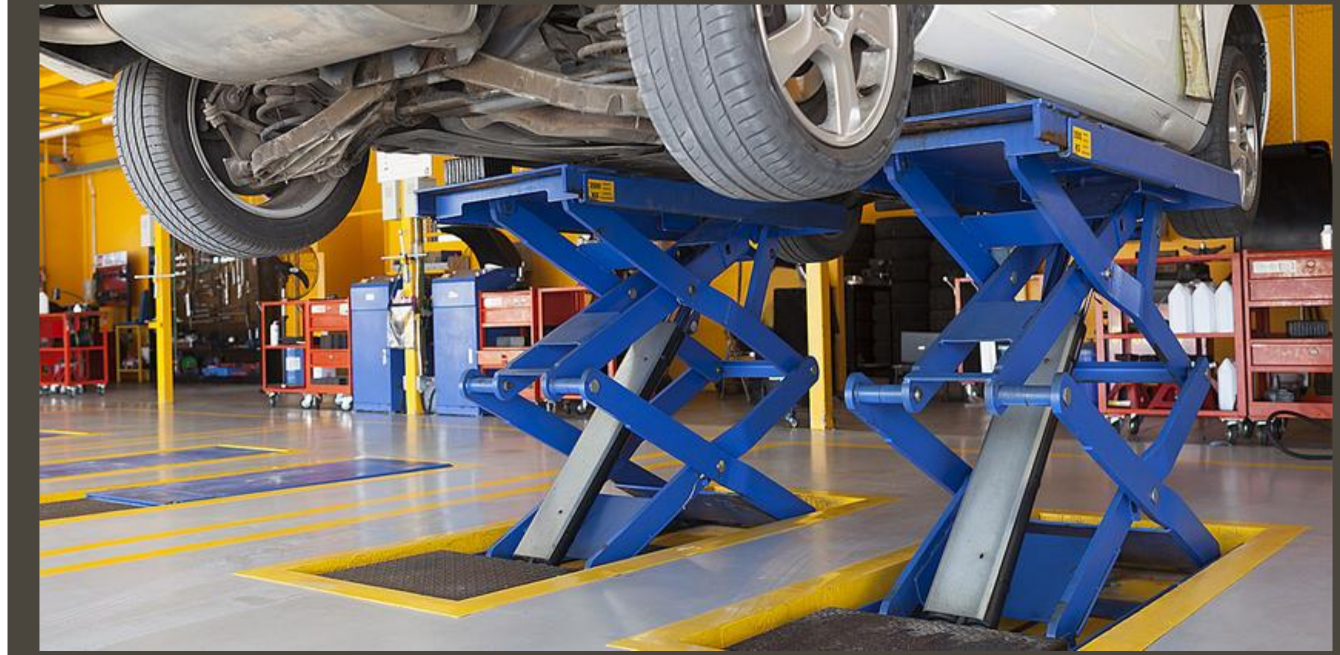
- **How it works:** An electric pump forces oil into a massive underground cylinder located directly beneath the elevator car.
- **The Physics:** As the fluid fills the cylinder, it pushes a heavy steel piston upward, smoothly lifting the elevator cab and all its passengers

2. Commercial and Public Spaces

Automotive Shop Lifts

When a mechanic needs to work underneath a vehicle, they drive it onto a platform and press a button, raising the car effortlessly above their head.

- **How it works:** This is a direct application of the classic U-shaped hydraulic lift. A small, motorized pump applies pressure to a small surface area of fluid, which transmits that exact same pressure to a massive piston under the car.



3. Everyday Tools and Conveniences

The Mechanic's Bottle Jack

If you've ever had to change a flat tire, you might have used a small, heavy iron tool called a bottle jack.

- **How it works:** By manually pumping a small lever up and down, you push a tiny piston into a fluid reservoir.
- **The Physics:** Because you are moving a small piston over a long distance, the fluid response allows you to lift one corner of a heavy SUV with just one hand

3. Everyday Tools and Conveniences

Office Chairs (Gas/Fluid Cylinders)

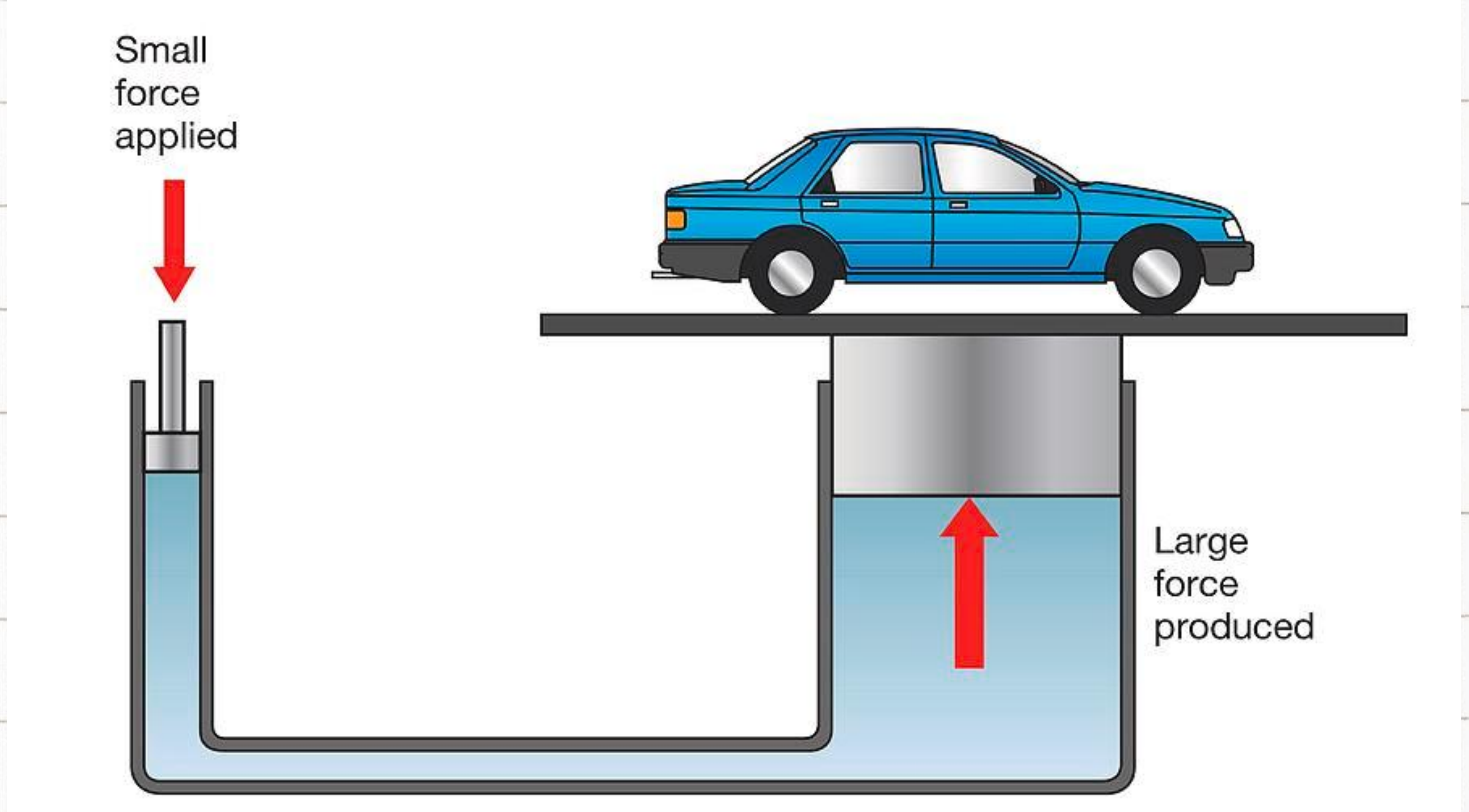
When you pull the lever under your desk chair to adjust its height, you are interacting with a simplified hydraulic/pneumatic cylinder.

- **How it works:** The lever opens a valve, allowing fluid to flow and adjust to your weight, locking smoothly into place when you release the lever.

GENERALIZATION

Pascal's principle states that any change in pressure applied to an enclosed, incompressible fluid is transmitted undiminished throughout the fluid in all directions.

A hydraulic lift uses liquid to turn a small push into a giant lift.

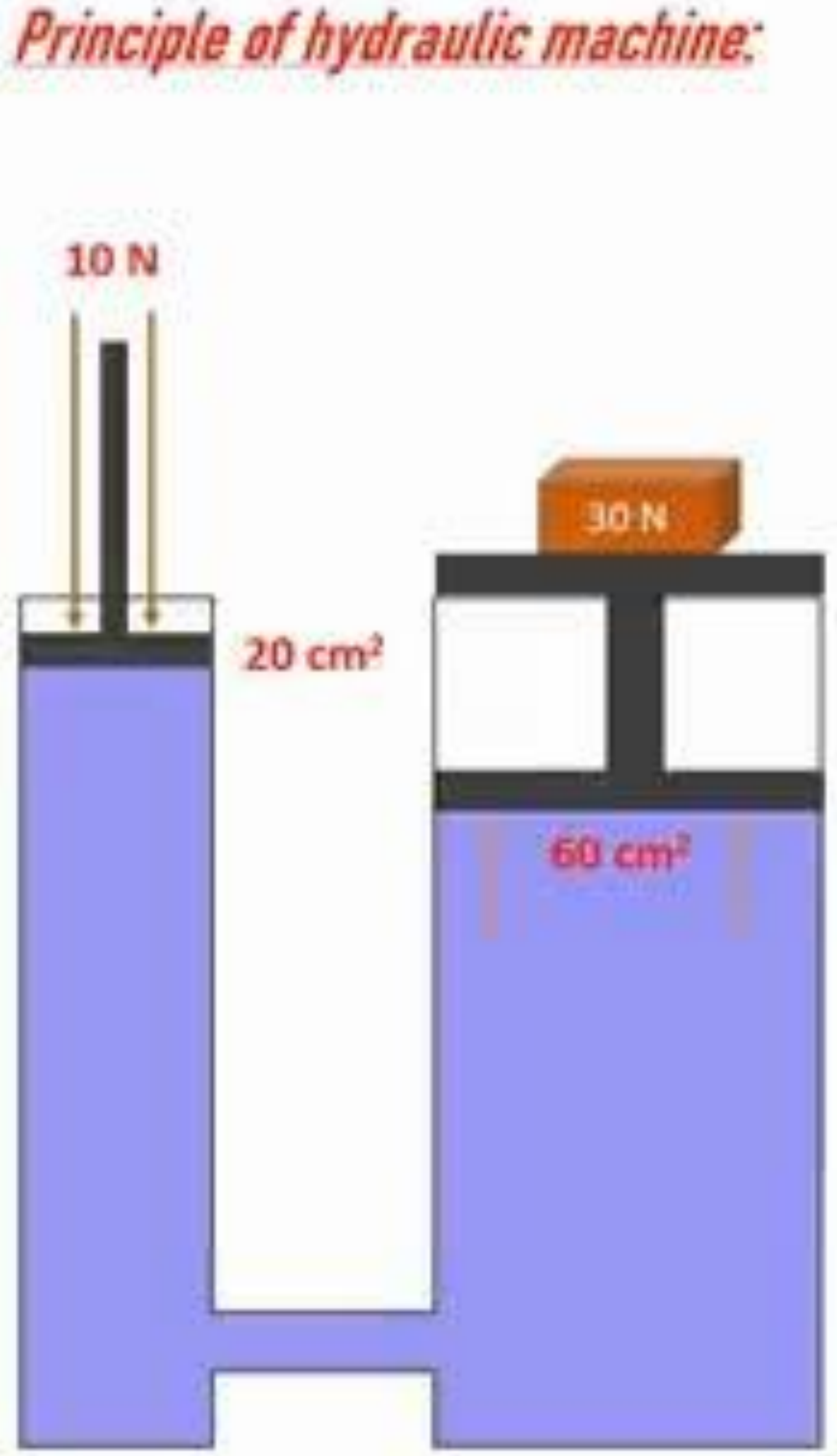


GENERALIZATION

Here is how it works using Pascal's principle:

1. Liquid can't be squished

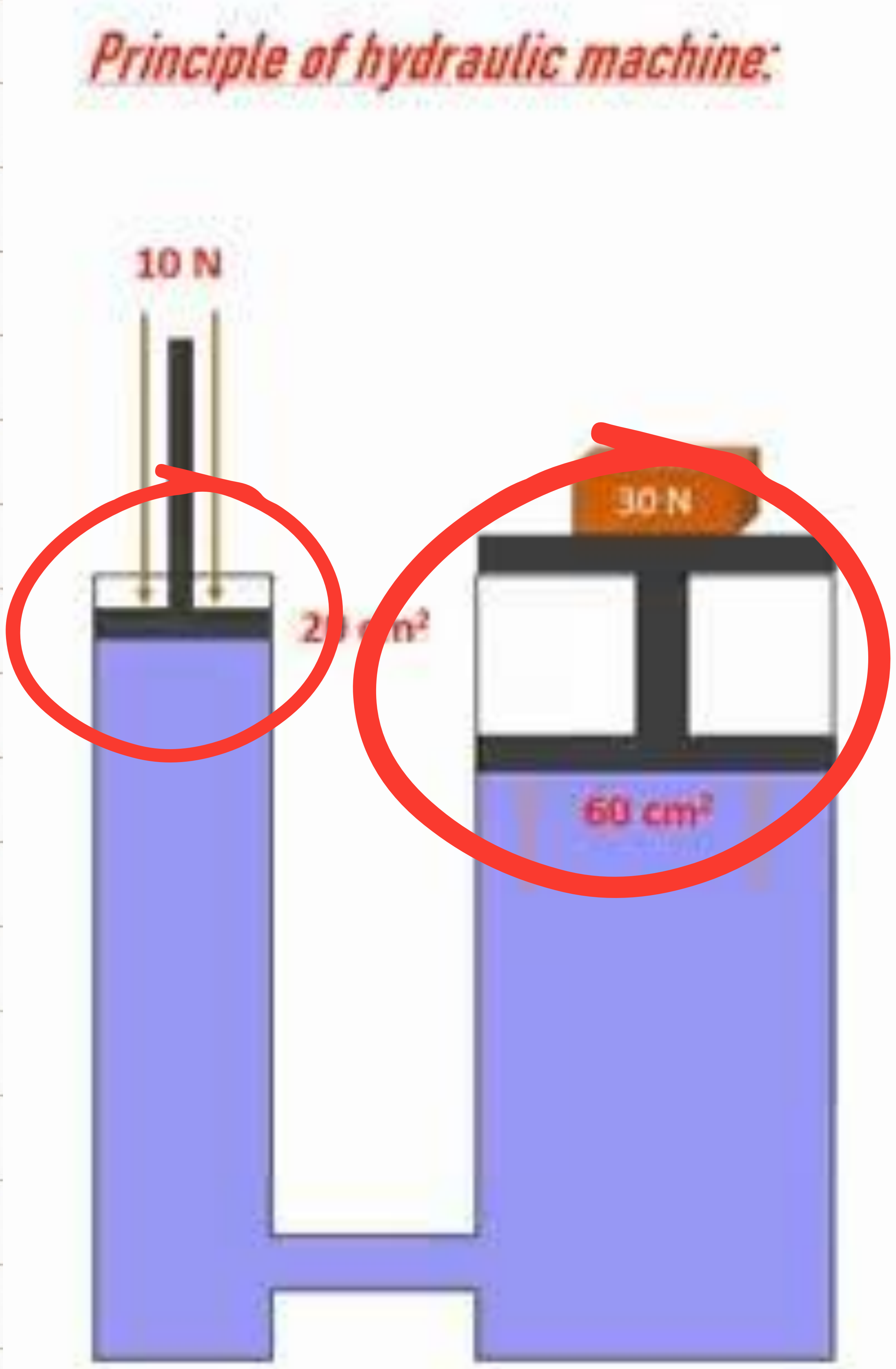
Unlike air, you cannot compress a liquid (like oil or water). If you push down on a liquid in a sealed pipe, that push travels through the fluid and presses against every single inch of the container with the exact same strength. This is Pascal's principle.



GENERALIZATION

2. The "Size Swap" Trick

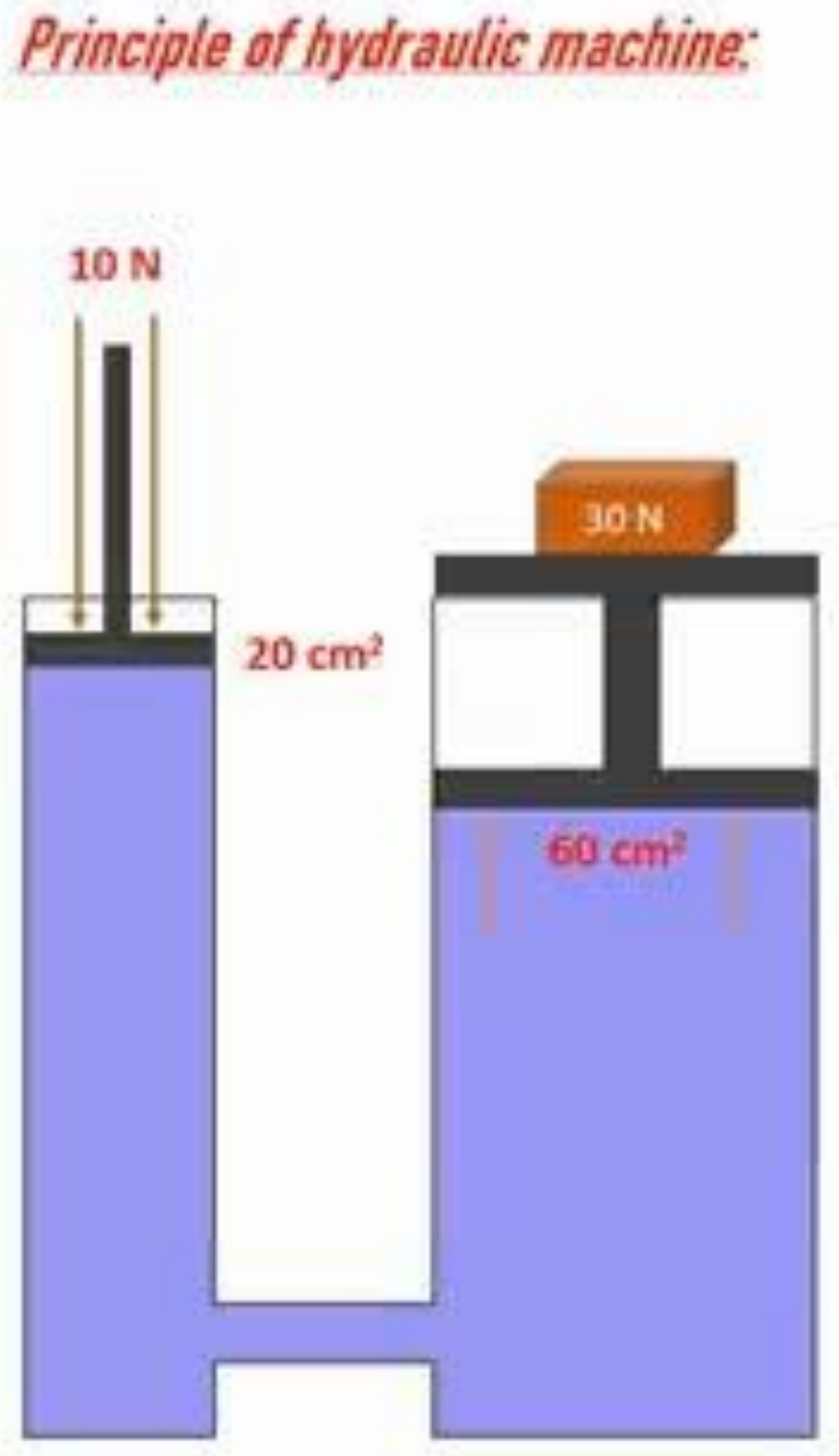
A hydraulic lift connects a small pipe to a giant pipe. You push down on the small side. Because it's small, it's easy to push. The liquid pushes that energy over to the giant side. Because the giant side has way more surface area, the liquid presses against it in thousands of places at once. All those tiny presses add up, creating a massive upward force capable of lifting a car.



GENERALIZATION

REMEMBER:
You don't get something for nothing. To lift a heavy car up by just 1 inch on the giant side, you have to push the small side down by 10 or 20 inches.

Summary: You swap distance for power. You push a long way with a weak force, and the lift translates that into a short way with a superpower force.



1. If you apply force to a trapped fluid, what happens to the pressure inside the container?

A) It disappears completely after a few seconds.

B) It spreads out equally to every part of the fluid.

C) It stays only directly beneath where you pushed.

D) It gets weaker the further it travels from your hand.

2. Which of the following everyday items relies on a hydraulic system to work?

A) A car’s braking system

B) A battery-powered flashlight

C) A standard desktop computer mouse

D) An electric toaster

3. What is the official unit used by scientists to measure pressure?

A) Newton (*N*)

B) Meter (*m*)

C) Kilogram (kg)

D) Pascal (*Pa*)

4. Why do hydraulic systems use liquid (like oil) instead of gases (like air)?

A) Liquids can be easily compressed into a smaller space.

B) Liquids are incompressible and pass force along instantly.

C) Liquids are lighter than gases.

D) Liquids make the machines move faster.

5. In a hydraulic lift, if you want to lift a very heavy load with a small input force, how should the output piston compare to the input piston?

A) The output piston must have a much smaller surface area.

B) Both piston must be the exact same size.

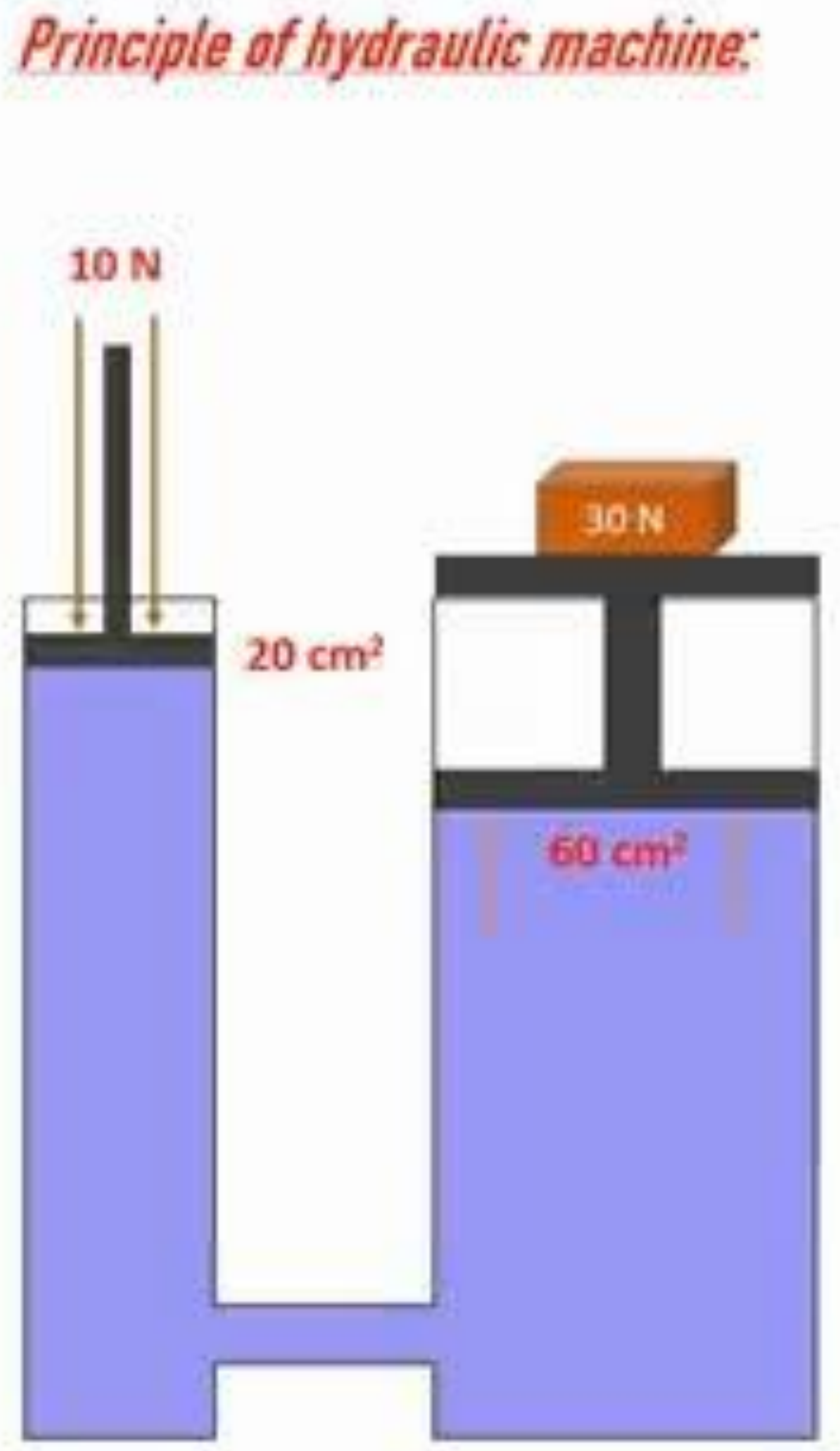
C) The output piston must have a much larger surface area.

D) The output piston must be filled with air instead of oil.

ACTIVITY: Hydraulic Prototype

Choose one hydraulic application from the following:

Hydraulic Robotic Arm
Hydraulic Lift
Hydraulic Excavator
Hydraulic Bridge
Hydraulic Crane
Any teacher-approved hydraulic device



ACTIVITY: Hydraulic Prototype

DEMONSTRATION

Each group will:

1. **Demonstrate how the prototype works.**
2. **Explain how pressure is transmitted through the fluid.**
3. **Identify the real-life application of their prototype.**
4. **Discuss challenges encountered and how they were solved.**

ACTIVITY: Hydraulic Prototype

Guide Questions

1. What hydraulic device did your group create?
2. How does your prototype demonstrate Pascal's Principle?
3. What happens when force is applied to one syringe?
4. How is pressure transmitted through the liquid?
5. What improvements would you make if given more time?